

Bayesian Decision Theory

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Outline

- 1 Classification
- 2 Losses & Risks
- 3 Discriminant Functions
- 4 Association Rules

Bernoulli Distribution

- Random variable $X \in 0, 1$
- Bernoulli: $P\{X = 1\} = p_0^X (1 - p_0)^{(1-X)}$
- Given a sample $\mathcal{X} = \{x^t\}_{t=1}^N$
- we can estimate $\hat{p}_0 = \frac{\sum_t x^t}{N}$

Classification

- Input $\vec{x} = [x_1, x_2]$, Output $C \in \{0, 1\}$
- Prediction:

$$\text{choose } \begin{cases} C = 1 & \text{if } P(C = 1|\vec{x}) > 0.5 \\ C = 0 & \text{otherwise} \end{cases}$$

- Equivalently:

$$\text{choose } \begin{cases} C = 1 & \text{if } P(C = 1|\vec{x}) > P(C = 0|\vec{x}) \\ C = 0 & \text{otherwise} \end{cases}$$

- E.g., Credit scoring
 - inputs are income and savings
 - Output is low-risk versus high-risk

Bayes' Rule

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- $P(\vec{x})$: *evidence*
 - If we ignore the classes, how like are we to see a value $\vec{x}$?

Bayes' Rule - Multiple Classes

$$\begin{aligned}P(C_i|\vec{x}) &= \frac{P(C_i)p(\vec{x}|C_i)}{p(\vec{x})} \\&= \frac{P(C_i)p(\vec{x}|C_i)}{\sum_{k=1}^K p(\vec{x}|C_k)P(C_k)}\end{aligned}$$

- $P(C_i) \geq 0$ and $\sum_{i=1}^K P(C_i) = 1$
- choose C_i if $P(C_i|\vec{x}) = \max_k P(C_k|\vec{x})$

Unequal Risks

- In many situations, different actions carry different potential gains and costs.
- Actions: α_j
- Let λ_{ik} denote the *loss* incurred by taking action α_j when the current state is actually in C_k
- *Expected risk* of taking action α_j :

$$R(\alpha_j|\vec{x}) = \sum_{k=1}^K \lambda_{ik} P(C_k|\vec{x})$$

- This is simply the expected value of the loss function given that we have chosen α_j
- Choose α_j if $R(\alpha_j|\vec{x}) = \min_k R(\alpha_k|\vec{x})$

Special Case: Equal Risks

- Suppose $\lambda_{ik} = \begin{cases} 0 & \text{if } i = k \\ 1 & \text{if } i \neq k \end{cases}$
- Expected risk of taking action α_i :

$$\begin{aligned} R(\alpha_i|\vec{x}) &= \sum_{k=1}^K K \lambda_{ik} P(C_k|\vec{x}) \\ &= \sum_{k \neq i} P(C_k|\vec{x}) \\ &= (1 - P(C_i|\vec{x})) \end{aligned}$$

- Choose α_i if $R(\alpha_i|\vec{x}) = \min_k R(\alpha_k|\vec{x})$
 - which happens when $P(C_i|\vec{x})$ is largest
- So if all actions have equal cost, choose the action for the most probable class.

Special Case: Indecision

- Suppose that making the wrong decision is more expensive than making no decision at all (i.e., falling back to some other procedure)
 - Introduce a special *reject* action α_{K+1} that denotes the decision to not select a “real” action
 - Cost of a reject is λ , $0 < \lambda < 1$

$$\lambda_{ik} = \begin{cases} 0 & \text{if } i = k \\ \lambda & \text{if } i = K + 1 \\ 1 & \text{if } i \neq k \end{cases}$$

The Risk of Indecision

- Risk:

$$\begin{aligned} R(\alpha_{K+1}|\vec{x}) &= \sum_{k=1}^K \lambda P(C_k|\vec{x}) \\ &= \lambda \\ R(\alpha_i|\vec{x}) &= \sum_{k \neq i}^K P(C_k|\vec{x}) \\ &= 1 - P(C_i|\vec{x}) \end{aligned}$$

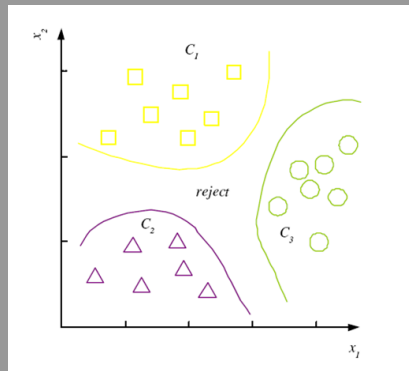
- Choose α_i if $P(C_i|\vec{x}) > P(C_k|\vec{x}) \forall k \neq i$ and $P(C_i|\vec{x}) > 1 - \lambda$
 - otherwise reject all actions

Discriminant Functions

An alternate vision.

- Instead of searching for the most probable class
- we seek a set of functions that divide the space into K decision regions $\mathcal{R}_1, \dots, \mathcal{R}_K$

$$\mathcal{R}_i = \left\{ \vec{x} \mid g_i(\vec{x}) = \max_k g_k(\vec{x}) \right\}$$



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- Allows us to use them when we have no info of the underlying distribution
- Later techniques will seek discriminant functions directly.

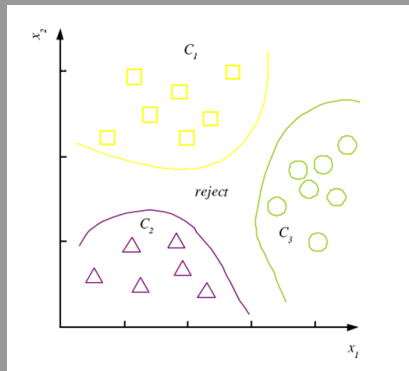
Bayes Classifier as Discriminant Functions

- We can form a discriminant function for the Bayes classifier very simply:

$$g_i(\vec{x}) = -R(\alpha_i|\vec{x})$$

- If we have a constant loss function, we can use

$$\begin{aligned} g_i(\vec{x}) &= P(C_i|\vec{x}) \\ &= \frac{P(C_i)p(\vec{x}|C_i)}{p(\vec{x})} \end{aligned}$$

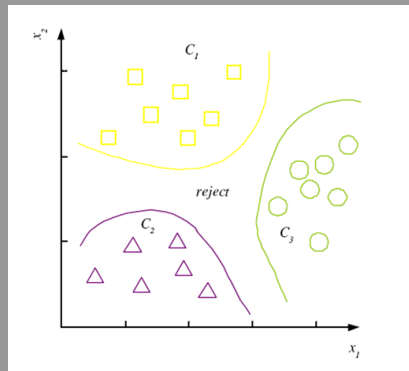


Bayes Classifier as Discriminant Functions (cont.)

$$g_i(\vec{x}) = \frac{P(C_i)p(\vec{x}|C_i)}{p(\vec{x})}$$

Because all the g_i above would have the same denominator, we could alternatively do:

$$g_i(\vec{x}) = P(C_i)p(\vec{x}|C_i)$$



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- $\text{Lift}(X \rightarrow Y) \equiv \frac{P(X,Y)}{P(X)P(Y)} = \frac{P(Y|X)}{P(Y)}$
- Support and confidence are more common
- Support and lift are symmetric

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- We want to find rules with support and confidence above designated thresholds

Agarwal's Apriori Algorithm

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- 2 Then find an association rule $\{X_i\}_{i \neq k} \rightarrow X_k$, using the values in each such tuple, that has high confidence

Apriori Algorithm - Subsets

- ① Start by finding $\{X_i\}$ with high support
 - $\text{Support}(\{X_i\} \rightarrow Y) \equiv P(X_1, X_2, \dots)$
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Apriori Algorithm - Subsets

- ① Start by finding $\{X_i\}$ with high support
 - $\text{Support}(\{X_i\} \rightarrow Y) \equiv P(X_1, X_2, \dots)$
 - If $\{X_i\}$ has high support, then all subsets of it must have high support.
- ② Then find an association rule $\{X_i\}_{i \neq k} \rightarrow X_k$, using the values in each such tuple, that has high confidence

Apriori Algorithm - Finding Support

$$\text{Support}(\{X_i\} \rightarrow Y) \equiv P(X_1, X_2, \dots)$$

- ① Start with “frequent” (high support) single items
- ② By induction,
 - ① Given a set of frequent k -item sets,
 - ② Construct all candidate $k + 1$ -item sets
 - ③ Make a pass evaluating the support of these candidates, discarding low-support sets.
- ③ Continue until no new sets discovered or until all inputs are gathered into one set.

Apriori Algorithm - Finding Confidence

$$\text{Confidence}(\{X_i\} \rightarrow Y) = \frac{P(X_1, X_2, \dots, Y)}{P(X_1, X_2, \dots)}$$

Given a k -item set with high support:

- 1 Try each possible single-consequent rule.

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 - ② **Construct all candidates with $j + 1$ consequents**

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 - ③ Discard those with low confidence
- ③ Repeat until no higher-consequent rules are found